Guitar Playing Robot

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Abstract

Robotic devices capable of playing musical instruments have been given much attention over the past decade, though devices capable of playing actual guitars have been met with limited success. The aim of this project was to construct a device capable of playing an actual unmodified acoustic guitar in a manner that most closely represented the way in which an actual human guitarist plays. An existing prototype, capable of playing only a single guitar string, was used as a starting point. Many methods of both plucking a string and actuating positions along the fret board of the guitar were researched. A solution was selected for achieving both of these functions and was consequently designed and constructed. The final prototype was actuated by numerous solenoids and a motor. Control was achieved via an Altera FPGA device and proved very successful at achieving the desired functionality. Song information can be input in the standard MIDI format and subsequently downloaded to the device and played at a reasonable tempo. The use of a very capable and flexible FPGA controlling device has provided ample room for expanding the current functionality of the robot in a wide range of respects.

Introduction

This project was conducted as a 2006 final year Electrical Engineering thesis project at the Electrical & Computer Systems Engineering department of Monash University. The objective of the project involved taking an existing guitar playing robot and upgrading the robot so as to be able to play over all six strings of an actual unmodified acoustic guitar.

Background

A previous robot was provided which came with severe limitations in its ability to play music; the main issue being that it was capable of playing only one string. The mechanical device used to actuate this one string made expansion to play additional strings impossible given the space constraints of an actual guitar. Thus, it was decided that a complete overhaul of the device would be required. Novel methods of plucking a string were researched and discussed.

Various devices have been created in the past in an attempt to robotically actuate stringed instruments, such as the guitar. The most notable of these are the:

- CrazyJ robot (Lawrence, Howard and Knueven 2000)
- LEMUR Guitarbot (Singer, Larke and Bianciardi 2003),
- Afasia: one man multimedia band (Jordan 2002), and
- the Taito guitar robot (Inglis 2004).

The Crazy J robot design is the most similar to what we were trying to achieve and it was the only guitar robot we discovered capable of playing an actual unmodified acoustic guitar. It consisted of a frame above an actual acoustic guitar where there were picks and fingertips that plucked and depressed the strings. This robot was very successful in its ability to play a variety of notes at a decent tempo. It was limited by the fact that the fingertips at the fretting end of the guitar were stationary, covering positions over only the first 4 frets. This limited the robot's playing note range. There were 24 separate fingertips which were spread out over 6 strings and 4 frets. Solenoids were located beside the guitar to actuate the fingertips via an intricate lever mechanism.

Plucking was achieved by using a pick on a thin vertical shaft. Bidirectional picking of the strings was achieved through the use of two solenoids per string.

Technical Details/Methodology

Our project was split up into several main components: plucking mechanism, fretting mechanism, movable carriage, electronics, onboard software and PC software.

The following section details the design process that was used to arrive at the final prototype.

Mechanics - Plucking Mechanism

As shown in Figure 1, the plucking mechanism on the existing guitar used a motor to spin a plectrum across a single string of the guitar. This device was very bulky and could not be used to fit within the confines of a 6 string playing mechanism.

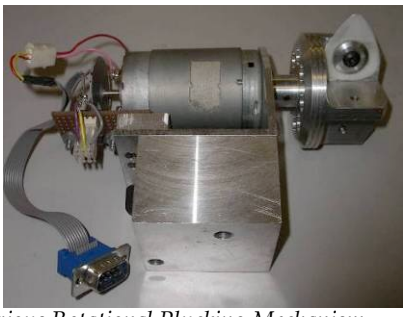


Figure 1.Previous Rotational Plucking Mechanism

Upon researching methods of exciting a string, it was decided that given the size constraint, the best method of excitation would be to use a harpsichord style plucking mechanism, similar to that used on the CrazyJ.

Actual harpsichord jacks were sourced to provide actuation of the string. A solenoid and lever mechanism was designed in order to move the plectrum up and down through the line of the string. Six identical devices were created such that each string could be actuated individually and at similar amplitudes of oscillation. The final prototype of the device is shown in Figure 2.

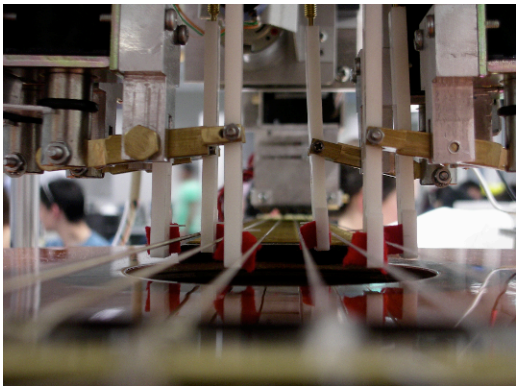


Figure 2. Harpsichord Plucking Mechanism

This device worked extremely well in achieving the desired functionality. Each string could be plucked individually at speeds up to approximately 9Hz.

The use of harpsichord jacks to excite the string produced sounds very similar to the sound created by a human guitarist's plucking action.

Another benefit of the new design was that on the return of the jack to the up position, the plectrum would not pluck the string; rather it would bend out of the way to bring a damper cloth into position. These damper cloths provided the capability of providing "note-off" functionality.

Mechanics - Fretting Mechanism

The most complicated part of a human guitarist's playing action is the work done to actuate differing positions on the fret board. Not only does a human guitarist depress one string at a time to play certain individual notes, but by depressing certain combinations of notes over the six strings of a guitar, the player can achieve a wide range of chords.

On researching what had previously been done in the area of mechanical actuation of the fret board, it was found that the majority of solutions involved a large static array of pneumatic or electromagnetic solenoids. This can be seen in such devices as the CrazyJ Guitar Playing Robot, the Taito Robot Guitar and Sergi Jorda's Afasia project.

This solution was well beyond our project budget due to the large amount of requisite solenoids and the serious mechanical challenges inherent in fitting that many solenoids within the limited available space.

A device capable of moving up and down the neck of the guitar with an array of "fingers" covering three frets was chosen to perform actuation of the fret board. This was done as it was achievable within the project budget and closely approximated what a human guitarist actually does while playing the guitar.

By modification of the available finger positions on the movable carriage, a range of chords could be achieved. An array of "fingers" was designed capable of playing all of the basic chords a guitarist would desire to play whilst minimising the number of solenoids required to achieve this. This array of solenoids is shown in Figure 3.



Figure 3. Minimalist Finger Array

In order to achieve actuation of these points, in the requisite space, small high-force solenoids were required. In order to minimise the power requirements of the device, magnetically latched solenoids were chosen with a magnetic holding force of 480g. This force was empirically determined as being adequate for holding down guitar strings.

These solenoids were arranged in a staggered way such that they all fitted onto the movable carriage. As shown in Figure 4, they were attached to levers on a number of combined pivot pins.

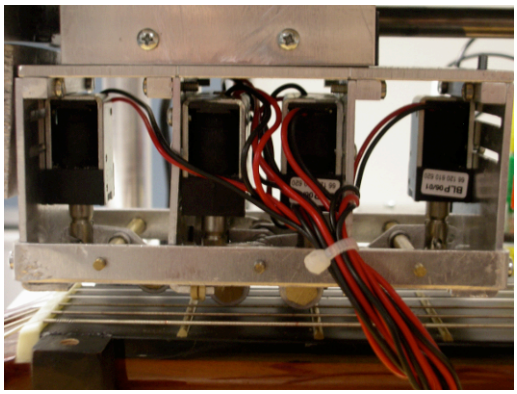


Figure 4. Fretting Mechanism

Mechanics - Movable Carriage

The fretting mechanism was required to move to different fret positions to play different notes and chords. This was achieved by attaching the carriage to a corkscrew thread salvaged from an old inkjet printer. As shown in Figure 5, a DC motor, salvaged from the same printer, drove the corkscrew thread.

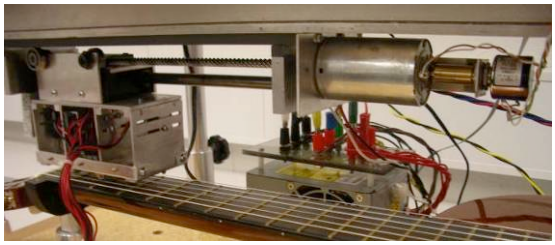


Figure 5. Movable Carriage Mechanism

Position feedback from the movable carriage was provided by a 10-turn potentiometer coupled to the drive shaft of the motor.

Electronics

The design of the electronics of the device had the following requirements:

- drive the magnetically latched fretting solenoids in either direction,
- drive the movable carriage motor in either direction at a variable speed,
- drive the plucking solenoids with a suitable amount of current, and
- provide feedback of the movable carriage position using sensor information.

These functions were separated into individual printed circuit boards (PCB) with a central breakout board from the Altera FPGA controller providing input/output (I/O) to these separate boards. For example, Figure 6 shows the solenoid driver PCB used to drive the fretting solenoids to hold down strings of the guitar.

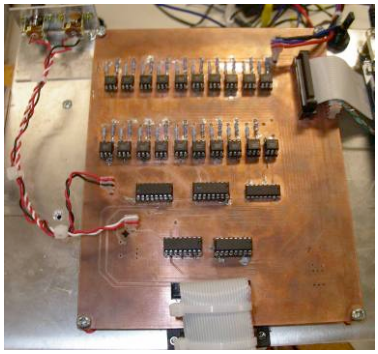


Figure 6. Fretting solenoid driver PCB

Figure 7 and Figure 8 show the circuitry used to drive the numerous solenoids of the robot.

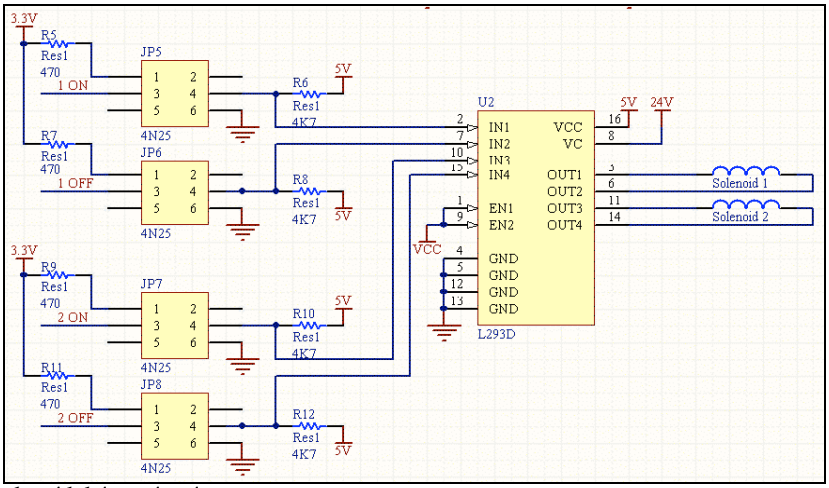


Figure 7. Fretting solenoid driver circuit

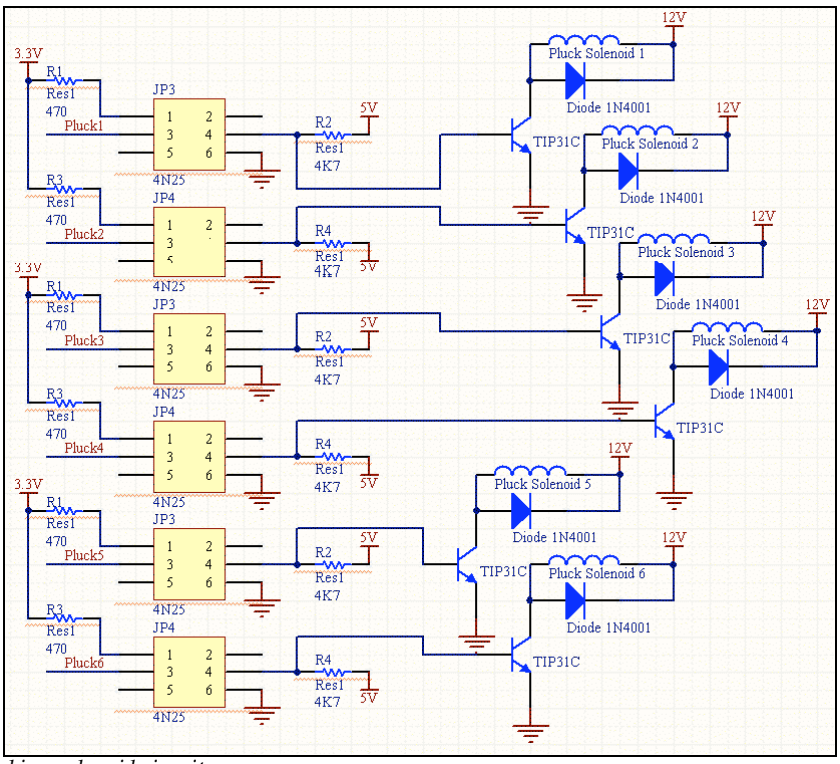


Figure 8. Plucking solenoid circuit

Fretting electronics

Fretting was achieved using a similar mechanism to that used for plucking except that it is done to a smaller scale and with magnetic latching pull-type solenoids. These solenoids require both a +24V supply to pull the solenoid in and a -24V supply to drop the solenoid out again. A small elastic band was used to assist the solenoid pin to drop back out again, when required.

As a bi-directional motor driver was used to control the direction of current, two digital outputs from the Altera microcontroller were used for each solenoid. All of these microcontroller outputs were opto-isolated by an optocoupler circuit. The full circuit of one fretting solenoid is shown in Figure 7.

Plucking electronics

Referring to Figure 8, each plucking solenoid was controlled by a single digital output signal from the Altera microcontroller. The 3.3 V output of the microcontroller was converted to a 5 V TTL level via an optocoupler. This voltage level was then subsequently converted to the 12 V needed by the solenoid by a motor driver IC.

Software

The software written to control the device consisted of two distinct parts:

Extraction Software - PC software for decoding type 0 and type 1 musical instrument digital interface (MIDI) files and writing their song data into a control information format easily utilised by the robot, and

Robot Control Software - Altera FPGA NIOS II microcontroller software used to run through PC extracted song data and send out the relevant commands to drive the robot in time with the song information.

PC extraction software

MIDI extraction software was developed for the project that extracts note information from a MIDI file and converts it into a format the robot is able to read. It does this by grouping the MIDI notes into chords, and then assigning a string to each note of the chord. A bridge position is then calculated depending on the array of available solenoids.

The program subsequently creates an output stream which, when read by the Altera microcontroller DE2 development board (Altera 2006), sends the necessary note on and off information as well as motor control information to play the guitar.

This output stream can be sent either to an external file or out the PC serial port to the Altera microcontroller memory.

Robot control microcontroller software

From the point of view of the robot, the initial header bytes of the song data obtained from the PC MIDI extraction software contains timing information for the song (pulses per quarter note and tempo) and the length of the song data.

Song information is then presented to the robot in a format consisting of a range of sequential chronological commands. The possible commands understood by the robot are:

- Note On: energises a finger solenoid or moves the finger carriage into position,
- Note Off: de-energises a finger solenoid,
- Pluck: plucks a string on the guitar,
- Dampen: dampens a string on the guitar, and
- Delay: waits for a prescribed amount of time.

Precise timing of these distinct tasks performed by the robot was programmed utilising a port of the MicroC real-time kernel on the NIOS soft-core microcontroller.

The device was programmed such that a central "control" task runs through the data stream byte-by-byte, posting messages into message-boxes as it proceeds. These message-boxes are then accessed by their corresponding relevant task so as to ensure the robot performs the desired function.

A collaboration graph of how this was achieved is shown in Figure 9.

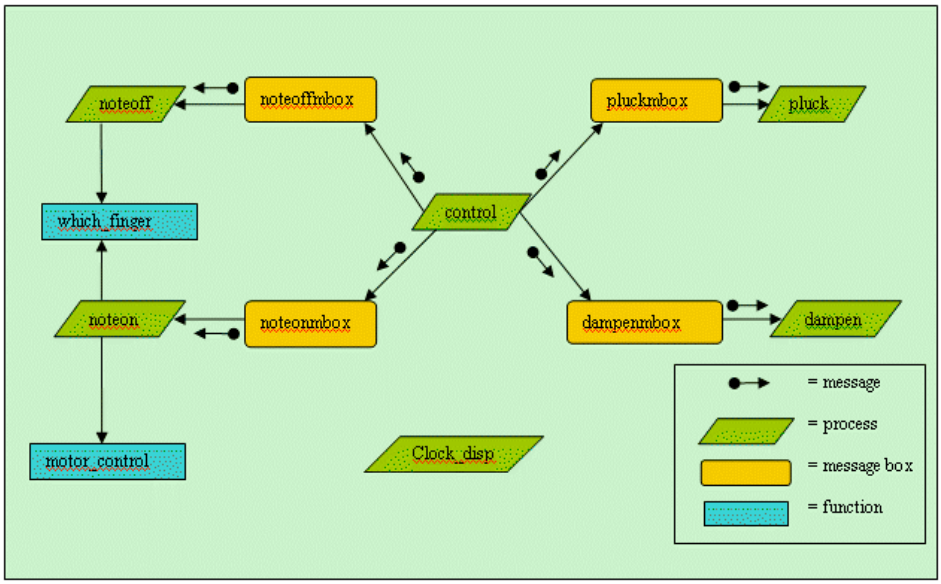


Figure 9. Real-time Collaboration Graph

Results

The plucking mechanism developed was found to perform well; strings were plucked in a timely fashion at a reasonable tempo.

For the fingering mechanism, it proved difficult to get all of the solenoids effectively holding down their respective strings. Once in place, however, acceptable operation was obtained.

Results of the position control system are shown below in Figure 10. The thin pink line indicates the measured position whereas the thick blue line represents the intended position. As shown in Figure 10, the position control was reasonably accurate. In the real world however, even the slightest error in the position can have a dramatic effect on the pitch of the It note played. The positioning control, therefore, was not perfect and could use further calibration. The final prototype is shown in Figure 11.

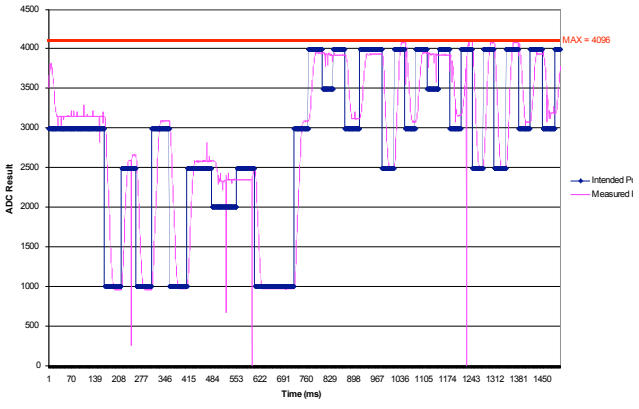


Figure 10. Position Control Results

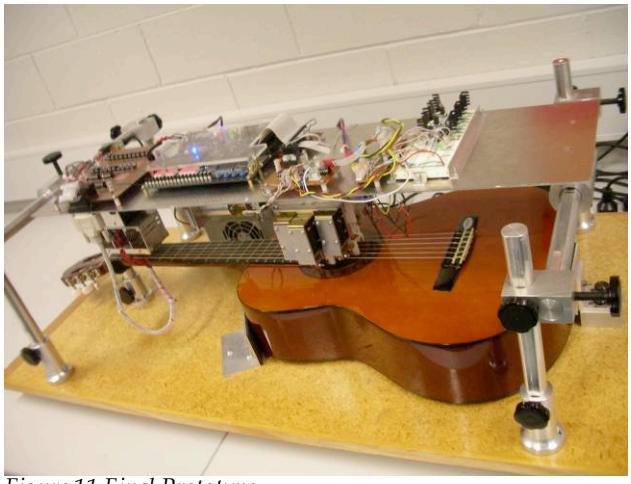


Figure 11 Final Prototype

Discussion

The robot has inherent limitations in its design that make it unable to play a wide range of music. These limits include (1) speed of movement and, (2) unplayable chords, and (3) unplayable notes.

The main speed limitation present in the device is the time it takes to get the movable fretting carriage into position. With the current position system, this can take anywhere from 100 - 200ms. This delay is offset against subsequent delays in the song by the MIDI extraction software, but it can still be too long to keep up with some song data.

Movement of the fretting carriage also generates noticeable mechanical noise which can interfere with the music being produced by the guitar. One way this could be overcome in future, however, is to use a suitable pickup inside the guitar, which would allow amplification of its tones. In this way, the desired guitar tones could be subsequently amplified to a level where the background fretting carriage mechanical noise becomes insignificant.

There are also a few problems with the setup of the robot as any minor adjustments to various heights or angles of the arrangement can cause the robot to fail to play correctly. This may require a good 30 minutes or more of adjustment to the robot in order for it to play properly and in tune. This can cause the robot to become frustrating to work with and is an area that still requires extensive investigation in an attempt to rectify it.

The fretting mechanism also needs to be expanded by the addition of more solenoids and improved by the use of a rubber material to hold down the strings.

The project could be improved in many areas, the main ones including:

Addition of more solenoids

The current project shows that it is possible to fit solenoids side by side in an array over all six strings. A future development could be to add more solenoids to the array so that every string and fret position is covered. This would allow more demanding songs to be played.

Improved fretting solenoid driver circuit

A major improvement to the solenoid driver circuit would be to modify it so that a different voltage is applied to allow the solenoid to drop out again. According to the data sheet, if a correct negative voltage is placed on it, there is a minimum force required to drop it out again.

Improved plucking mechanism

The current plucking mechanism severely limits the range of expression possible with the guitar robot. For example, playing at differing volume levels, i.e. with different dynamics, is currently not possible. The timbre of the plucking cannot also be varied. Different plucking mechanisms could be investigated which would better facilitate this.

Improved Manufacturing

The manufacturing process could be simplified and improved by remaking a lot of what was made. Parts could be made more accurately as there is now more information about the correct placement of all the parts and how they will work together.

Improvements to both the PC and Altera software

Currently, both programs work quite well depending on the type of input MIDI file but there are problems when some MIDI files are used. These problems are mainly due to the fret position found by the algorithm and also the relative sluggishness of the motor driver compared to what is sometimes required of it.

Conclusion

The resulting robot has achieved most of the project design objectives; being capable of (1) playing over all six strings of the guitar, and (2) playing a range of chords. Using a PC, the robot can accept standard MIDI files as input. These are downloaded to it and subsequently control its playing. Some simple songs have since been demonstrated on the robot, including Deep Purple's "Smoke On The Water", "Twinkle Twinkle Little Star" and AC/DC's "Back In Black". It differs from previous designs by the use of the movable fretting carriage as well as the use of a more flexible and capable FPGA controlling device.

With only minor improvements to its existing operation (eg. addition of more solenoids), a significantly broader range of chords could be achieved together with a more flexible arrangement for playing notes faster. In addition, given the use of a FPGA for this application, this has left the door open for significantly increasing the capabilities of the guitar robot. For example, given the large number of free I/O pins still available on this device, together with its significant processing capability, it would be quite feasible to control two or even three guitar robots with the one FPGA. Therefore, MIDI files comprising musical pieces meant for two or three guitars could be downloaded to the one FPGA, which would then control these different guitars to perform these multi-part pieces.

Real-time digital signal processing could also be performed by such a FPGA, either in hardware or software or a combination of the two. Therefore, it would be possible for the FPGA to perform a Fast Fourier transform on the sounds produced by the guitar, analyse the spectrum produced and detect, for example, the guitar going out of tune.

Acknowledgements

Many thanks must go out to the team at the Monash University Electrical Engineering workshop, notably Tony Brosinsky and Maurice Gay. They worked tirelessly to help us to complete this project.

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